**7** **(a)** State Lenz's law of electromagnetic induction.

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.....

..... [2]

**(b)** A helicopter hovering in stationary equilibrium has four rotors, each of length 12 m, as shown in the view from above in Fig. 7.1.

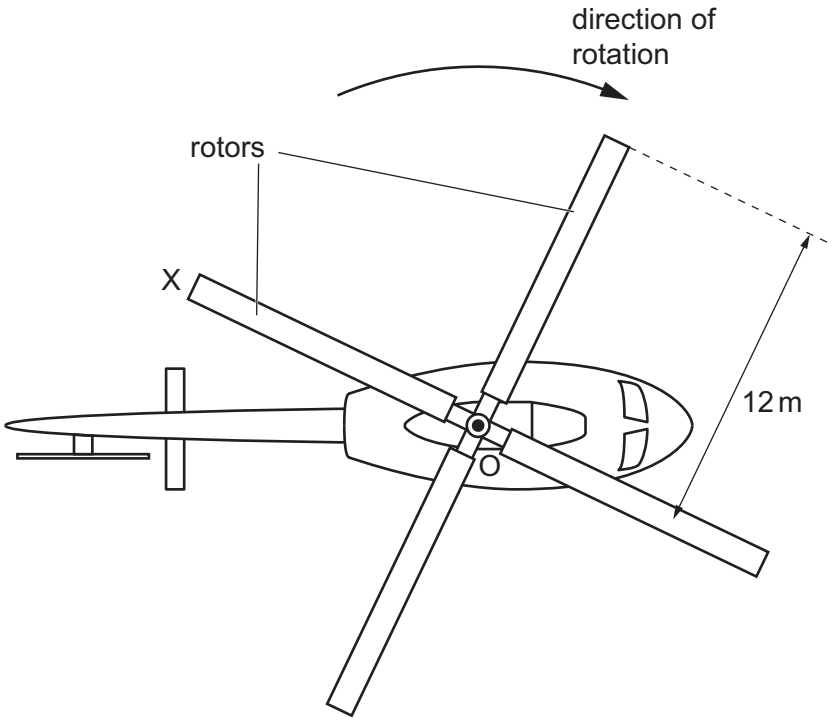



**Fig. 7.1**

The vertical component of the Earth's magnetic field at the helicopter is downwards with a flux density of 0.047 mT.

The rotors each rotate in a horizontal plane in the direction shown with a frequency of 85 Hz.

**(i)** Calculate the magnetic flux $\Phi$ cut by rotor OX during one complete rotation. Give a unit with your answer.

$\Phi$ = ..... unit ..... [3]

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**(ii)** Determine the magnitude of the electromotive force (e.m.f.) induced across the length of rotor OX.

e.m.f. = ..... V [2]

**(iii)** Use Lenz's law to explain whether end O or end X of the rotor is at the higher potential.

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..... [2]

[Total: 9]

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